

Summer Math Adventure

W1
D1

Week 1 · Day 1 | Place Value: Reading & Writing Big Numbers

Name: _____

Date: _____

PLACE VALUE

Remember: Every digit has a *place* AND a *value*.

In 4,728,013 — the digit **7** is in the **hundred-thousands** place. Its value is **700,000**.

Periods help us read numbers: | Billions | Millions | Thousands | Ones |

Part A – Name That Place

Write the **place** and **value** of the underlined digit.

1. 3,462,805

Place:

Value:

2. 27,914,600

Place:

Value:

3. 513,002,744

Place:

Value:

4. 8,004,371

Place:

Value:

Part B – Expanded Form

Write each number in expanded form. Example: $4,302 = 4,000 + 300 + 2$

5. 605,814 =

6. 3,070,009 =

7. Write the number for: $400,000 + 60,000 + 500 + 7 =$

Part C – Comparing Numbers

Write $<$, $>$, or $=$ in each circle.

8. 3,841,002 3,841,020

9. 70,500,000 7,500,000

10. 999,999 1,000,000

11. 24,060,000 24,006,000

Name: _____ Date: _____

ROUNDING & ESTIMATION

Rounding Rule: Look at the digit ONE place to the RIGHT of where you're rounding.

- If it's 0–4 → round *down* (keep the digit the same, replace the rest with zeros)
- If it's 5–9 → round *up* (add 1 to the digit, replace the rest with zeros)

Part A – Round It!

Number	Nearest 10	Nearest 100	Nearest 1,000
4,763			
28,549			
305,281			
1,999,500			

Part B – Estimation in Action

Estimate by rounding each number to its **leading digit**, then compute. Show both the estimate AND the exact answer.

1. A school has 4,812 students and 3,276 moved in over summer. About how many students total?
Estimate:

Exact:

2. A factory makes 78,340 bolts per day. About how many bolts in 5 days?
Estimate:

Exact:

3. Phoenix gets about 8 inches of rain per year. Estimate rain over 12 years.
Estimate:

Exact:

Part C – Is the Estimate Reasonable?

Circle **YES** or **NO** and explain why.

4. Mia says $6,489 + 3,211 \approx 10,000$. YES / NO
Why?

Summer Math Adventure

W1
D3

Week 1 · Day 3 | Addition & Subtraction: Large Numbers

Name: _____

Date: _____

ADDITION & SUBTRACTION

Key Reminder: Always line up digits by *place value* before adding or subtracting.

When subtracting, if the top digit is smaller, you must **regroup (borrow)** from the next column.

Check: Addition answer – one addend = other addend. Subtraction answer + subtracted number = starting number.

Part A – Solve It (Show all work in the space)

1. $485,372 + 246,918 =$

Answer:

2. $3,004,000 - 1,867,425 =$

Answer:

3. $72,308 + 9,847 + 140,003 =$

Answer:

4. $10,000,000 - 4,382,619 =$

Answer:

Part B – Error Hunt

Each problem below has a mistake. Find it, explain it, and write the correct answer.

5. A student solved: $500,000 - 238,471 = 272,439$

Correct answer:

What went wrong?

6. A student solved: $304,072 + 58,900 = 362,072$

Correct answer:

What went wrong?

Summer Math Adventure

W1
D4

Week 1 · Day 4 | Multiplication: Patterns & Properties

Name: _____

Date: _____

MULTIPLICATION FOUNDATIONS

Properties to remember:

- **Commutative:** $6 \times 9 = 9 \times 6$
- **Associative:** $(2 \times 3) \times 5 = 2 \times (3 \times 5)$
- **Distributive:** $7 \times 13 = 7 \times 10 + 7 \times 3 = 70 + 21 = 91$
- **Multiplying by 10, 100, 1,000:** Just add zeros! $47 \times 100 = 4,700$

Part A – Powers of 10 Patterns

1. $83 \times 10 =$

$83 \times 100 =$

$83 \times 1,000 =$

2. $7 \times 40 =$

$7 \times 400 =$

$70 \times 400 =$

Part B – Distributive Property

Break apart each problem and solve. Show all steps.

3. 6×47
 $= 6 \times (__ + __)$
 $= __ + __$
 $=$

4. 8×93
 $= 8 \times (__ + __)$
 $= __ + __$
 $=$

5. 5×108
 $= 5 \times (__ + __)$
 $= __ + __$
 $=$

6. 12×25
 $= 12 \times (__ + __)$
 $= __ + __$
 $=$

Part C – Multi-Digit Multiplication

7. $346 \times 7 =$

8. $524 \times 38 =$

9. $2,004 \times 6 =$

Part D – Word Problem (Cumulative)

Summer Math Booklet · Grade 4 · W1 · Week 1 · Day 4

Multiplication Foundations

10. Destiny's school uses 48 boxes of pencils each month. Each box has 144 pencils. The school year has 9 months. How many pencils does the school use in a year? First, estimate to find the answer.

Summer Math Adventure

W1
D5

Week 1 · Day 5 | Division: Meaning, Remainders & Word Problems

Name: _____

Date: _____

DIVISION

Division = splitting into equal groups.

$372 \div 6$: "How many groups of 6 fit in 372?"

Remainder: the leftover amount that can't form a full group.

Check: (quotient $\times$ divisor) + remainder = dividend $\rightarrow (62 \times 6) + 0 = 372 \checkmark$

Part A – Division with Remainders

1. $97 \div 4 =$

Q:

R:

2. $253 \div 7 =$

Q:

R:

3. $8,145 \div 9 =$

Q:

R:

Part B – What Does the Remainder Mean?

For each problem, decide: Do you round UP, round DOWN, or USE THE REMAINDER?

4. 75 kids need to take the school van. Each van holds 8 kids. How many vans are needed?

Work: $75 \div 8 = \underline{\hspace{1cm}} \text{ R } \underline{\hspace{1cm}} \rightarrow$ Answer:

_____ vans (Why?

_____)

5. 130 cookies are packed equally into boxes of 12. How many *full* boxes can be packed?

Work: $130 \div 12 = \underline{\hspace{1cm}} \text{ R } \underline{\hspace{1cm}} \rightarrow$ Answer:

_____ full boxes (leftover cookies: $\underline{\hspace{1cm}}$)

6. 4 friends share \$47 equally. How much does each get, and how much is left over?

Each gets: \$

_____ Leftover: \$

Week 1 Quiz – Number Sense & Operations

W1
QUIZ

Topics: Place Value · Rounding · Addition · Subtraction · Multiplication · Division

Name: _____

Date: _____

Score: _____ / 20

Show all work. Each question is worth 1 point unless marked otherwise.

Section 1 – Place Value & Rounding (5 pts)

1. What is the value of the digit 6 in 46,803,210?

2. Write $5,000,000 + 30,000 + 400 + 9$ as a standard number.

3. Round 8,372,649 to the nearest hundred-thousand.

4. Which is greatest? Circle one:
2,047,000 2,470,000 2,407,000

5. **(2 pts)** Write the number that is 100,000 MORE than 6,900,050.

Section 2 – Addition & Subtraction (5 pts)

6. $748,293 + 351,708 =$

7. $5,000,000 - 3,284,736 =$

8. **(3 pts)** A library has 2,403,811 books. They donate 385,200 and receive 740,000 new ones. How many books does the library have now?

Section 3 – Multiplication & Division (10 pts)

9. $463 \times 27 =$

10. $50 \times 700 =$

11. $3,276 \div 9 =$

12. $5,824 \div 16 =$

13. **(2 pts)** Use the distributive property to solve 14×35 . Show two steps.